

Transfer Pricing Data Transformation Case Study

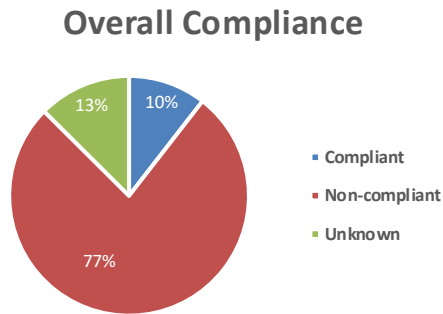
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1. Approach

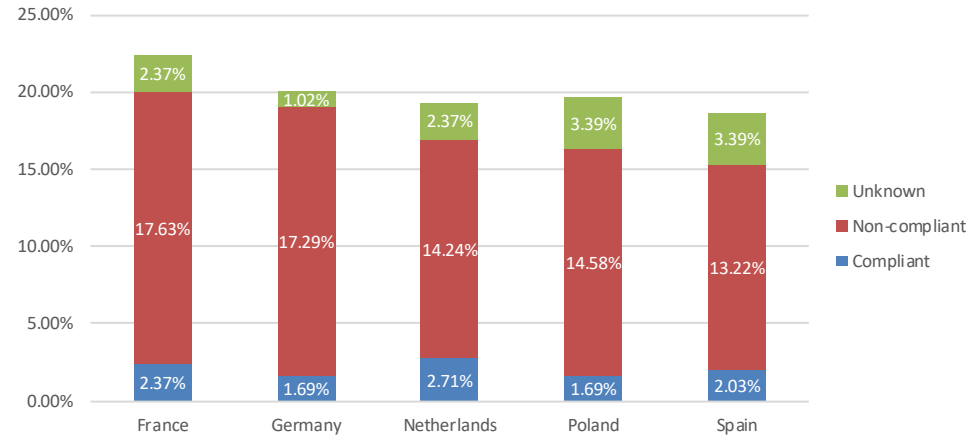
* Data preparation

Step	Assumption	Reasoning	Result
1. Removed duplicates	There are duplicate records in the dataset	Identified and removed fully identical records to prevent duplicate transactions and avoid inaccurate analysis	5 fully identical rows were deleted, partial matches are retained
2. Standardised Entities, Country, and Transaction type	Entity names, country and Transaction type may contain inconsistent formatting or extra spaces	Removed extra spaces and standardised entity names to ensure consistency and improve data accuracy for analysis	All Entity names, Country and Transaction type were cleaned and standardised
3. Standardised date format	Dates may have inconsistent formats across records	Converted all dates into a consistent YYYY-MM-DD format to enable accurate sorting, filtering, and analysis	All dates were formatted consistently as YYYY-MM-DD
4. Converted amount into EUR	Exchange rate: 1 USD = 0.9 EUR 1 PLN = 0.22 EUR	Transactions were recorded in different currencies. Converted transaction amounts into EUR to enable comparable transfer pricing analysis	All transaction amounts were converted into EUR using the defined exchange rates
5. Flagged missing data	Some records contain incomplete information	Identified missing values and flagged them instead of removing records to preserve data completeness while highlighting data quality issues	37 missing rows are flagged under Comment
6. Calculated Margin %	Calculated Margin % = (Amount - Cost Base) / Cost Base where possible	Applied the formula Margin % = (Amount - Cost Base) / Cost Base to evaluate profitability against arm's length benchmark ranges	Margin % was calculated for all records where sufficient data was available
7. Assessed compliance status	Arm's length benchmark range: - Goods: 3%-8% - Services: 5%-10% - Royalties: 8%-15%	Compared each transaction's calculated Margin % against the relevant arm's length benchmark range based on transaction type to determine whether it complies with transfer pricing requirements	Each transaction was classified as Compliant, Non-compliant, or Unknown

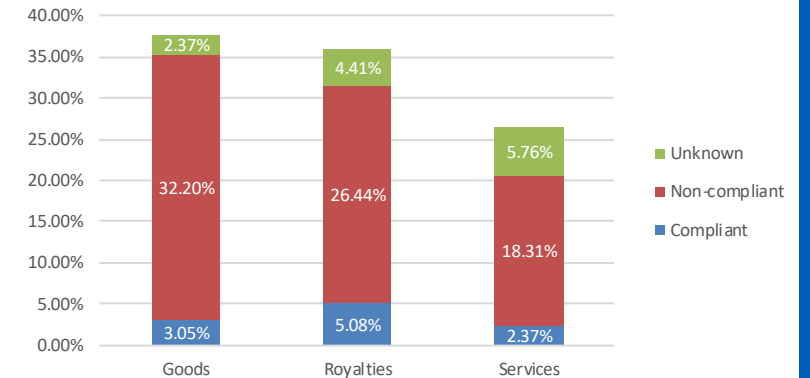
2. Key findings



Compliance by country



Compliance by Transaction Type



* Overall compliance risk

- Overall, only **10% of transaction are compliant** while **77% are classified as non-compliant** -> The high level of non-compliance suggest potential transfer pricing violation and audit risk, requiring additional review and escalations
- **13% of transactions have unknown** status -> Indicating data quality gaps and unreliable reporting, suggesting additional reviews are essential

* Compliance risk by country

- Most problematic countries are **France, Germany, and Poland**
 - **France** has the **highest non-compliance** share of approximately **17.7%**
 - The non-compliant rate of **Germany and Poland** follows at **17.29% and 14.58%** respectively, while these two nations have the **lowest compliance rate of below 2%**
 - **Poland**: the **highest Unknown** rate of 3.39% -> low data quality.
 - The Netherlands and Spain have lower non-compliance rates at 14,24% and 13,22%, compared to other countries
- => *France, Germany, and Poland look most problematic due to*
- + *Highest non-compliance (France)*
 - + *Worst mix of non-compliance and data quality (Germany, Poland)*

* Compliance risk by Transaction type

- **Goods** have the **highest non-compliance rate of 32.2%**, followed by Royalties with 26,44% while Services have a much lower non-compliance rate of 18.31%
-> concentration of risk in specific transaction types
- Services: highest Unknown rate of 5.76% -> poor transparency

3. Risks identified

*** Top 3 risks observed**

Priority	Risk	Evidence from the Dataset	Business Impact
1	High level of non-compliance	77% of transactions are classified as non-compliant	Significant compliance exposure, potential penalties, audit findings
2	Low data quality/ Missing data	13% of transactions have unknown status	Weak monitoring capability and difficulty identifying true risk exposure
3	Concentration of issues	Goods (32.2% non-compliant) and Royalties (26.44% non-compliant) represent largest problem areas	Indicates process gaps and concentrations of transactions

*** Likely root causes**

Data issues

- Incomplete transaction classification
- Missing supporting documentation
- Lack of automated validation rules

Process issues

- Missing or ineffective compliance checks in the end-to-end process
- Manual compliance checks can cause inconsistent outcome

Governance issues

- Different countries may have different policies
- Lack of standardised transfer pricing documentation procedures across countries

4. Recommendations

* Process improvement	1. Standardised templates:	Create global TP documentation templates for all entities and transaction types
	2. Validation rules:	Introduce automated checks for missing fields, incorrect classifications, and TP deviations
	3. Governance controls:	Establish ownership model, periodic reviews, escalation process, and compliance KPIs
* Automation ideas	1. Automated TP deviation flagging	- Identify transactions outside approved TP ranges and trigger review workflows - Automatically classify records as compliant / non-compliant / requiring review
	2. Dashboarding & reporting solution (Power BI)	- Real-time view of compliance status by country, entity, transaction type, and risk level - Update a matrix including countries, transaction type and risk level (high, medium, low – defined by non-compliance rate) to follow and control more easily
* AI Use Case	- Draft local files and master files - Identify inconsistencies across documents - Accelerate annual documentation updates - Detect unusual transaction patterns compared with historical behaviour - Predict transactions likely to become non-compliant - Prioritise high-risk cases for compliance teams	